

AI & ML For Malware Detection – CybAI Final Report

Author: MJeat

Publication Date: 17th August 2026

GitHub: <https://github.com/MJeat>

Introduction

Project Nickname: CybAI

VM OS: REMnux

Model Type: LightGBM, LogisticRegression

System Type: Basic Web UI Version

Scan Competence: Files and URLs

Main Features: 6 file types scanner, PE files, pre-defined reasoning

Project Detailed Summary

Dataset Usage

- PhiUSIIL Phishing URL Dataset
- Source: [Kaggle — PhiUSIIL Phishing URL Dataset](#)

Processing

- Selected only URLs and labels
- Removed missing URLs/labels
- Removed duplicate URLs
- Saved as `cleaned_url_list.csv`

Label Mapping

Class Labels

Label 1 corresponds to a legitimate URL; label 0 to a phishing URL

- 0 = Phishing
- 1 = Legitimate

Final Cleaned URL Dataset

235,370 URLs

EDA Findings

- 57.29% legitimate / 42.71% phishing → reasonably balanced
- Phishing URLs have a higher median length (34) than legitimate URLs (27)
- Phishing URLs have substantially larger length outliers, reaching 6,097 characters
- Keywords such as login, verify, account, secure, and update show strong class differences
- URL length and individual keywords provide useful signals but should not be used as standalone detectors

YARA Baseline

- Precision: 99.13%
- Recall: 11.10%
- F1-score: 19.97%

YARA produced 89,360 false negatives, demonstrating that signature/rule-based detection alone misses a large portion of phishing URLs.

ML Model – CybAI

A Logistic Regression classifier was trained using character-level TF-IDF features.

Raw URL > Character TF-IDF + 3–5 character n-grams > Logistic Regression > Phishing / Legitimate

The TF-IDF vectorizer was fitted only on the training set, while the test set was transformed using the learned vocabulary.

Initial ML Results

- Accuracy: 99.59%
- Precision: 100.00%
- Recall: 99.04%
- F1-score: 99.52%

The model substantially outperformed the YARA baseline on the random stratified test split. However, the unusually high performance requires further validation, particularly for **domain**

leakage and protocol-related shortcut learning, before treating it as representative of real-world performance.

DEMO

1. URL – Legitimate – Safe

LLM-For-Software-SecuriCybAI - Malware Scanner

http://localhost:5000/scan/url

CybAI - Malware ScannerScanLogin (Admin)

Scan Result for: https://www.apple.com

SAFE

Confidence: 99.91% | Detected by: cybai

Target Information

Name	https://www.apple.com
Type	URL

Layer 1 - URL/Domain Blacklist

Not in blacklist

Layer 2 - URL YARA Rules

Layer 3 - CybAI URL Model

Phishing Probability: 0.1%

Model Label: Legitimate

2. URL – Legitimate – Suspicious

LLM-For-Software-SecuriCybAI - Malware Scanner

http://localhost:5000/scan/url

CybAI - Malware ScannerScanLogin (Admin)

Scan Result for: https://en.wikipedia.org/wiki/Malware

SUSPICIOUS

Confidence: 62.15% | Detected by: cybai + yara

Target Information

Name	https://en.wikipedia.org/wiki/Malware
Type	URL

Layer 1 - URL/Domain Blacklist

Not in blacklist

Layer 2 - URL YARA Rules

Layer 3 - CybAI URL Model

Phishing Probability: 88.8%

Model Label: Phishing

88.8%

3. URL – Phishing – Suspicious/Malicious

The screenshot shows the CybAI Malware Scanner web interface. The browser address bar displays `http://localhost:5000/scan/url`. The page title is "CybAI - Malware Scanner". The main content area shows the scan result for the URL `http://login-paypal-secure.tk/verify`. A red banner at the top indicates the result is "MALICIOUS" with a confidence of 90.0% and detected by cybai + yara. Below this, the "Target Information" table lists the Name as `http://login-paypal-secure.tk/verify` and the Type as "URL". The "Layer 1 - URL/Domain Blacklist" section shows "Not in blacklist". The "Layer 2 - URL YARA Rules" section shows "2 rule(s) matched": `URL_Suspicious_Keywords` and `URL_Suspicious_Combination`. The "Layer 3 - CybAI URL Model" section shows a "Phishing Probability: 100.0%" and a "Model Label: Phishing". A red progress bar at the bottom indicates 100.0% completion.

Scan Result for: `http://login-paypal-secure.tk/verify`

MALICIOUS
Confidence: 90.0% | Detected by: cybai + yara

Target Information

Name	<code>http://login-paypal-secure.tk/verify</code>
Type	URL

Layer 1 - URL/Domain Blacklist

Not in blacklist

Layer 2 - URL YARA Rules

2 rule(s) matched

- URL_Suspicious_Keywords
- URL_Suspicious_Combination

Layer 3 - CybAI URL Model

Phishing Probability: **100.0%**

Model Label: **Phishing**

100.0%

4. File – Malware– Suspicious/Malicious

The screenshot shows the CybAI Malware Scanner web interface. The browser address bar displays `http://localhost:5000/scan/file`. The page title is "CybAI - Malware Scanner". The main content area shows the scan result for the file `ar15.mips`. A red banner at the top indicates the result is "MALICIOUS" with a confidence of 99.0% and detected by ml-ember2024 + yara. Below this, the "Target Information" table lists the Name as `ar15.mips`, Type as "ELF", SHA256 as `7479b14f9fe222c8e9917b58a9951a773330195bdc2da392e97eae3eb2e4ae12`, Family (combined) as "Malware.Url.Obfuscation", Family (YARA) as "Malware.Url.Obfuscation (via yara_rule_name:URL_Obfuscation)", and Family (EMBER2024) as "Unknown (via ml-unsupported-filetype)". The "Layer 1 - Hash Blacklist" section shows "Not in local_hashes.csv". The "Layer 2 - YARA Rules" section shows "6 rule(s) matched": `URL_Obfuscation`, `URL_Query_Parameters`, `URL_Suspicious_Combination`, `SUSP_ELF_LNX_UPX_Compressed_File`, and `domain`. The "Layer 3 - EMBER2024 ML" section shows an "ML Score: 100.0% - MALICIOUS (100.0%)" and "Top signals": `string_1` (malicious), `string_68` (malicious), and `string_106` (safe).

Scan Result for: `ar15.mips`

MALICIOUS
Confidence: 99.0% | Detected by: ml-ember2024 + yara

Target Information

Name	<code>ar15.mips</code>
Type	ELF
SHA256	<code>7479b14f9fe222c8e9917b58a9951a773330195bdc2da392e97eae3eb2e4ae12</code>
Family (combined)	Malware.Url.Obfuscation
Family (YARA)	Malware.Url.Obfuscation (via yara_rule_name:URL_Obfuscation)
Family (EMBER2024)	Unknown (via ml-unsupported-filetype)

Layer 1 - Hash Blacklist

Not in local_hashes.csv

Layer 2 - YARA Rules

6 rule(s) matched

- URL_Obfuscation
- URL_Query_Parameters
- URL_Suspicious_Combination
- SUSP_ELF_LNX_UPX_Compressed_File
- domain

Layer 3 - EMBER2024 ML

ML Score: 100.0% - MALICIOUS (100.0%)

Top signals:

- string_1 (malicious)
- string_68 (malicious)
- string_106 (safe)

5. File – Known Malware– Suspicious/Malicious

Scan Result for: *file*

MALICIOUS

Confidence: 100.0% | Detected by: hash-local_hashes.csv

Target Information

Name	file
Type	Win64
SHA256	72b8ee76e31527fa94e863ef15580037a2148bf59d6f981faab92a82f5403d58
Family (combined)	Malware.Ispe64
Family (YARA)	Unknown (via hash-match)
Family (EMBER2024)	Unknown (via hash-match)

Layer 1 – Hash Blacklist

Matched: local_hashes.csv – first found 2026-08-07

Layer 2 – YARA Rules

No matches

Layer 3 – EMBER2024 ML

skipped — hash match

Indices

1. EDA – Exploratory Data Analysis

To answer:

- How balanced is the dataset
- How long are phishing vs legitimate URLs?
- What characters appear frequently in each class?
- Are there obvious patterns or anomalies?
- Is the dataset suitable for our raw-URL ML model?

```
Phishing URLs:
count    100520.000000
mean      46.221439
std       61.181745
min       14.000000
25%       26.000000
50%       34.000000
75%       48.000000
max       6097.000000
Name: URLLength, dtype: float64
```

```
Legitimate URLs:
count    134850.000000
mean      27.228610
std       4.815612
min       16.000000
25%       24.000000
50%       27.000000
75%       30.000000
max       58.000000
Name: URLLength, dtype: float64
```

=====

URL LENGTH COMPARISON

=====

```
Phishing mean:    46.22
Legitimate mean:  27.23
Phishing median:  34.00
Legitimate median:27.00
```

=====

CHARACTER ANALYSIS

=====

```
'@' -> Phishing: 1.50% | Legitimate: 0.00%
'?' -> Phishing: 6.10% | Legitimate: 0.00%
'=' -> Phishing: 5.45% | Legitimate: 0.00%
'&' -> Phishing: 2.80% | Legitimate: 0.00%
'%' -> Phishing: 0.88% | Legitimate: 0.00%
'-' -> Phishing: 36.17% | Legitimate: 7.46%
'_' -> Phishing: 3.55% | Legitimate: 0.00%
'.' -> Phishing: 100.00% | Legitimate: 100.00%
'/' -> Phishing: 100.00% | Legitimate: 100.00%
```

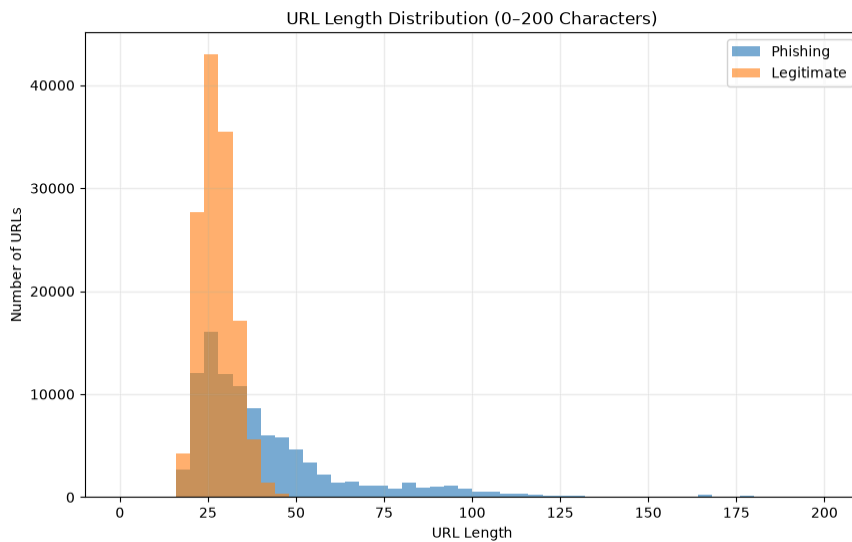
=====

KEYWORD ANALYSIS

=====

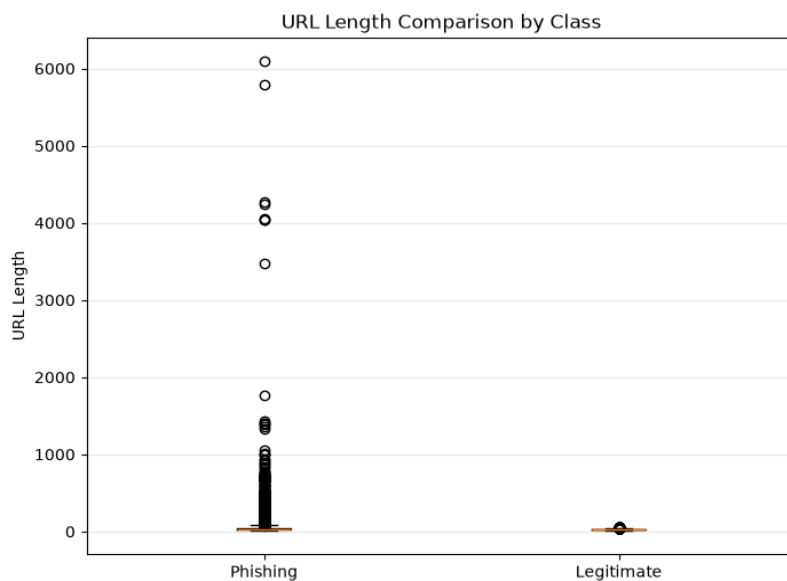
```
'login' -> Phishing: 3433 | Legitimate: 13
'verify' -> Phishing: 315 | Legitimate: 3
'account' -> Phishing: 1068 | Legitimate: 23
'secure' -> Phishing: 558 | Legitimate: 22
'update' -> Phishing: 840 | Legitimate: 21
'signin' -> Phishing: 886 | Legitimate: 9
'password' -> Phishing: 39 | Legitimate: 7
```

=====



Histogram Explanation:

- Legitimate URLs are heavily concentrated around 20–35 characters
- Phishing URLs have a much wider distribution
- Phishing URLs extend far beyond the legitimate range
- The extreme 6,097-character URLs are mostly not relevant to understanding the normal distribution



Box Plot Explanation:

- This is mainly to see the length of the URL
- It proves that a Phishing URL most likely contains longer characters.

2. TF-IDF + Character n-gram + Feature Extraction

```
from sklearn.feature_extraction.text import TfidfVectorizer
import pandas as pd

cleaned_url_df = pd.read_csv(r"D:\Phase-3-CybAI\Phase_3_phishing_detector_model\data\processed\cleaned_url_list.csv")

# Get number of URLs
extrac_url_col = cleaned_url_df["URL"]
print("The number of URL: ", len(extrac_url_col), "Rows")

# Get number of generated features (TF-IDF + n-grams)
vectorizer = TfidfVectorizer(ngram_range=(3,5),
                             analyzer="char",
                             min_df=5)  # min_df=5 means a feature must appear in at least 5 URLs to be kept

# Fit the data FIRST so the vectorizer can learn the vocabulary
"""
This is fine for this learning experiment, but when we build the actual ML model,
we must split the dataset before fitting the vectorizer to prevent test-set information from leaking into training.
"""

# This is when it becomes a compressed sparse format
X = vectorizer.fit_transform(extrac_url_col)

# Get the feature names
feature_names = vectorizer.get_feature_names_out()

# print("Get the feature names: ", feature_names)  # Should not print the entire thing. Not good practice.
print("Get the feature names: ", feature_names[:20])
print("Get the length of the feature names: ", len(feature_names))
```

3. Train/Test Split

```
=====
Total URLs: 235370 Rows
=====
X_train: 188296 (80.0%)
X_test: 47074 (20.0%)
=====
Total Training URLs: 188296
Total Testing URLs: 47074
=====
```



```

Label mapping:
0 = Phishing
1 = Legitimate
=====
Full Dataset Class Distribution:
label
1    57.29%
0    42.71%
Name: proportion, dtype: str
=====
y_train Class Distribution (80% Split):
label
1    57.29%
0    42.71%
Name: proportion, dtype: str

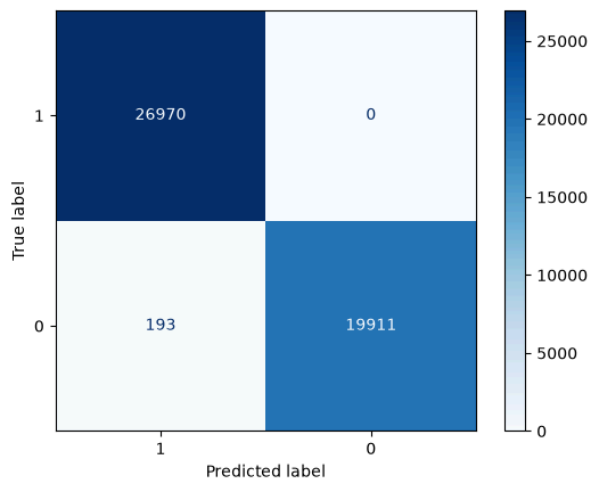
y_test Class Distribution (20% Split):
label
1    57.29%
0    42.71%
Name: proportion, dtype: str
=====
The shape of the X_train_tfidf (188296, 268327)
The shape of the X_test_tfidf (47074, 268327)
=====

```

4. Confusion Matrix

Matrix format for labels [1, 0], where 0 = Phishing (positive):

[[TN, FP], [FN, TP]]



Note:

I am aware of the zero detection for FN. More fine-tuning is needed and can be executed in the future.

5. Model Classification Report

MODEL PERFORMANCE

Accuracy: 0.9959
Precision: 1.0000
Recall: 0.9904
F1-score: 0.9952

CLASSIFICATION REPORT

	precision	recall	f1-score	support
Phishing (0)	1.00	0.99	1.00	20104
Legitimate (1)	0.99	1.00	1.00	26970
accuracy			1.00	47074
macro avg	1.00	1.00	1.00	47074
weighted avg	1.00	1.00	1.00	47074

Note:

I am aware of the high percentages of the model. However, for the sake of phase 3 and its definition, it is good enough. The CybAI project from phase 1 until phase 3 is now complete.

More fine-tuning is needed and can be executed in the future.